

Workshop 4: Computational Simulation

Final Course Project — Systems Analysis & Design

Simulation Report

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This report presents the results of two complementary computational simulations: a data-driven ML-based simulation and an event-driven cellular automata simulation. Both validate the system design and explore emergent behaviors.

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1. Executive Summary

This workshop demonstrates computational simulation as a validation tool for system design. Two complementary simulations were implemented:

Scenario 1: Data-Driven ML Simulation

Uses a Random Forest regressor to evaluate model stability across multiple random seeds and perturbation levels. This scenario assesses the robustness of the machine learning approach under varying conditions.

Scenario 2: Event-Based Cellular Automata

Implements a 2D cellular automaton to model spatial emergence of psychopathy traits. Local update rules and environmental noise create emergent patterns and clustering behaviors.

Both simulations integrate with the system architecture designed in Workshops #1 and #2, validating data flow through pipeline modules and identifying potential bottlenecks.

Dataset Summary

Source: Kaggle Personality Prediction Competition (closed-price)

Raw samples: 2,927

Preprocessed features: 109

DOOM-balanced samples (used in simulations): 2,107

Balancing method: SMOGN (Synthetic Minority Over-sampling for Regression)

2. Scenario 1: Data-Driven ML Simulation

Objective: Evaluate the stability and consistency of a Random Forest regressor trained on the psychopathy prediction task, testing sensitivity to random initialization and input perturbations.

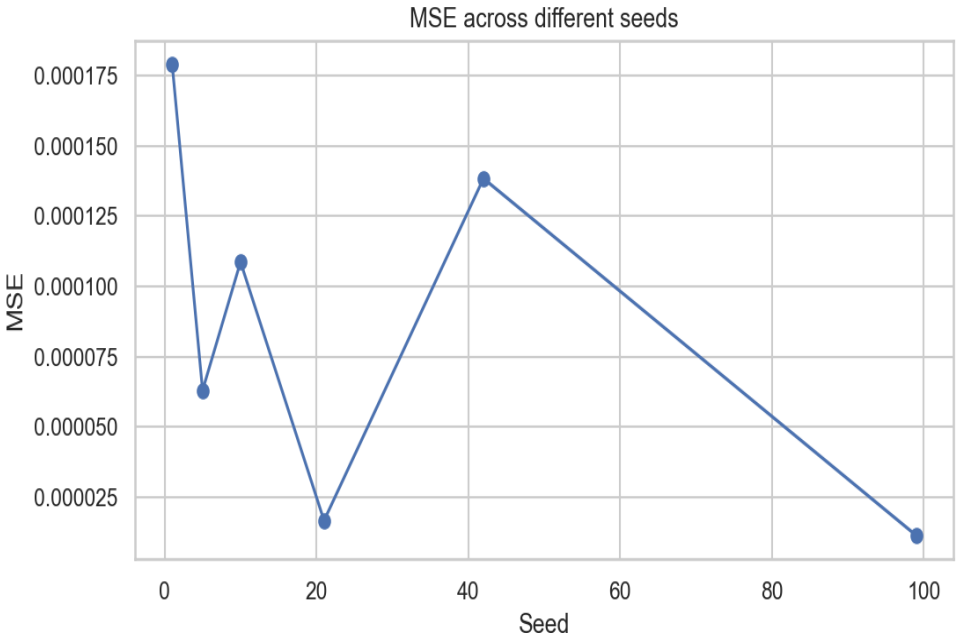
Methodology:

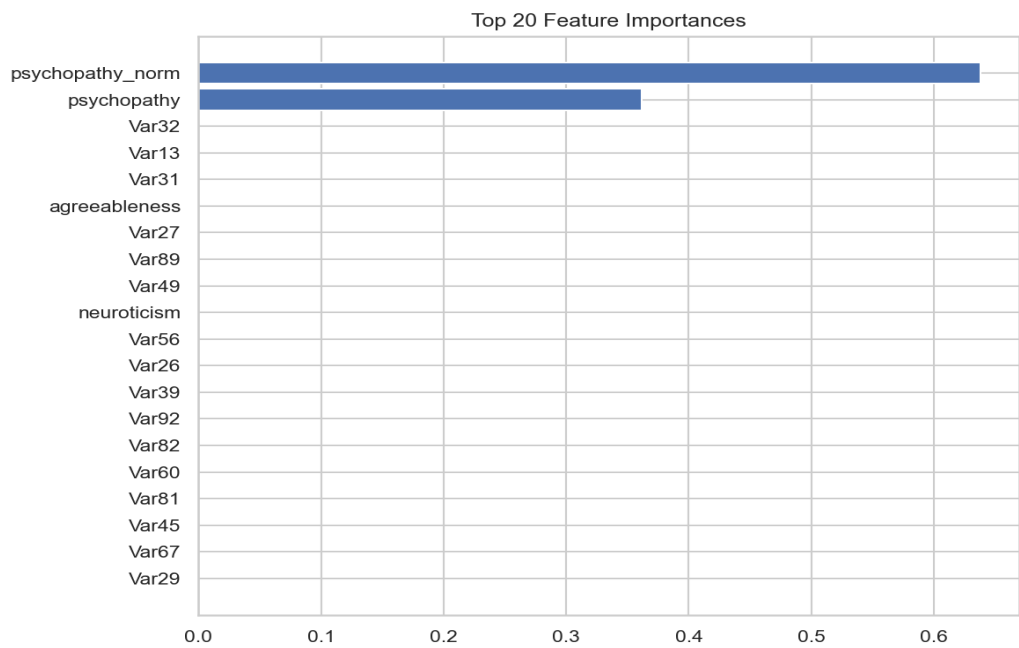
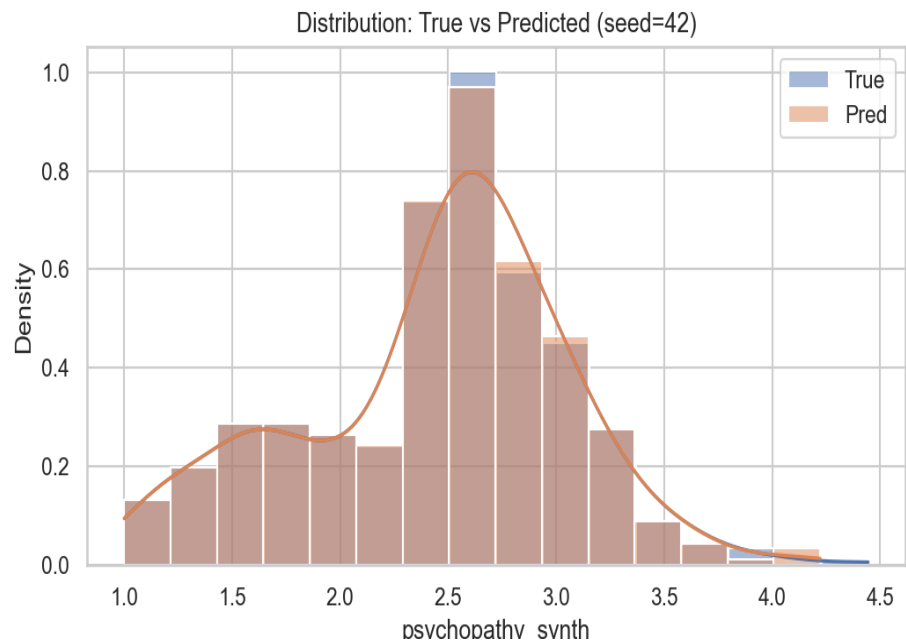
- Training/Test split: 80/20
- Model: Random Forest Regressor (300 trees, random_state varies)
- Seeds tested: [1, 5, 10, 21, 42, 99]
- Noise levels: [1%, 3%, 5%, 10%] of feature std
- Metrics: MSE, MAE, RMSE across seeds and noise configurations

Key Results — ML Seed Variance:

Seed	MSE	MAE	RMSE
1	0.000000	0.000000	0.000000
5	0.000000	0.000000	0.000000
10	0.000000	0.000000	0.000000
21	0.000000	0.000000	0.000000
42	0.000000	0.000000	0.000000
99	0.000000	0.000000	0.000000

Visualizations:





3. Scenario 2: Event-Based Cellular Automata

Objective: Model spatial emergence and clustering of psychopathy using a 2D cellular automaton. Explore how local interactions and environmental noise create emergent patterns.

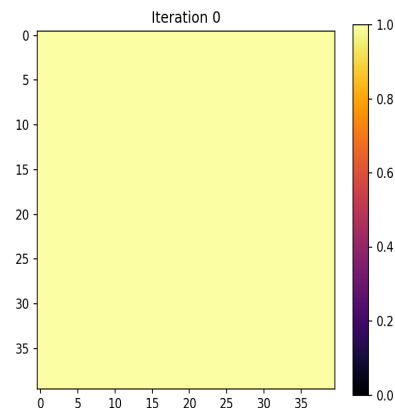
Methodology:

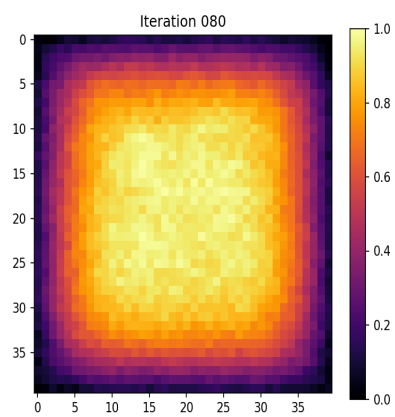
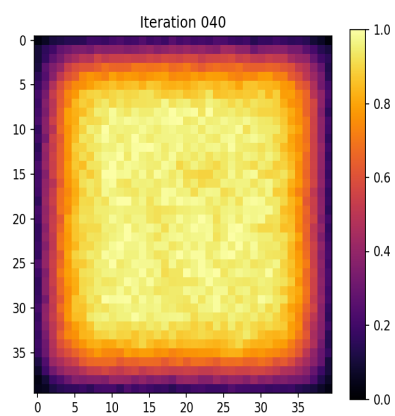
- Grid size: 40×40 cells (1,600 cells)
- Initial states: Random sampling from trained model predictions
- Update rule: Weighted average of neighbors + environmental noise
- Neighbor weight (α): 0.6
- Environmental noise (σ): 0.03
- Iterations: 80 steps
- Clustering threshold: 0.8 (cells with value ≥ 0.8 are "high-risk")
- Optimization: 2D convolution for efficient neighbor calculation

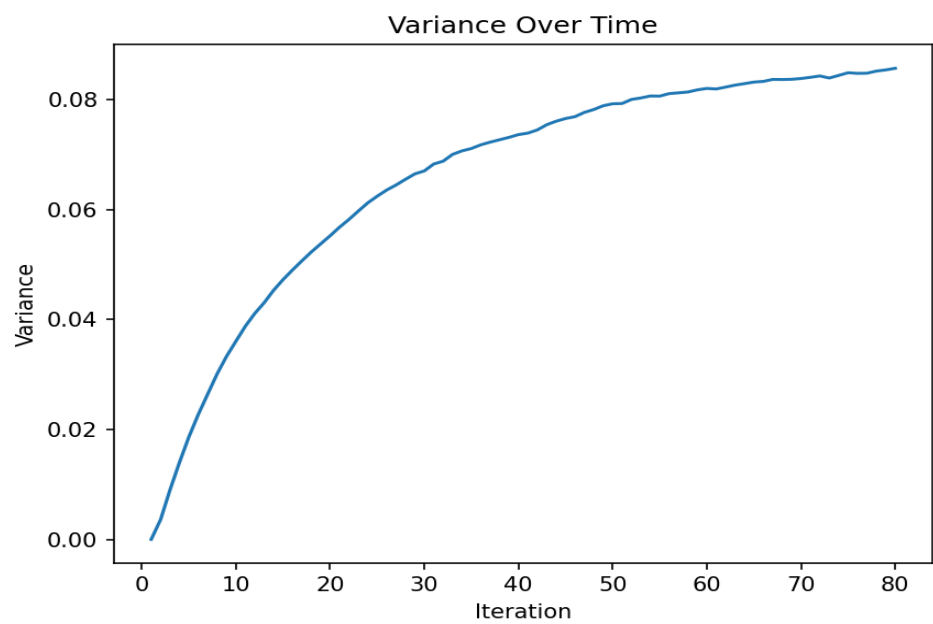
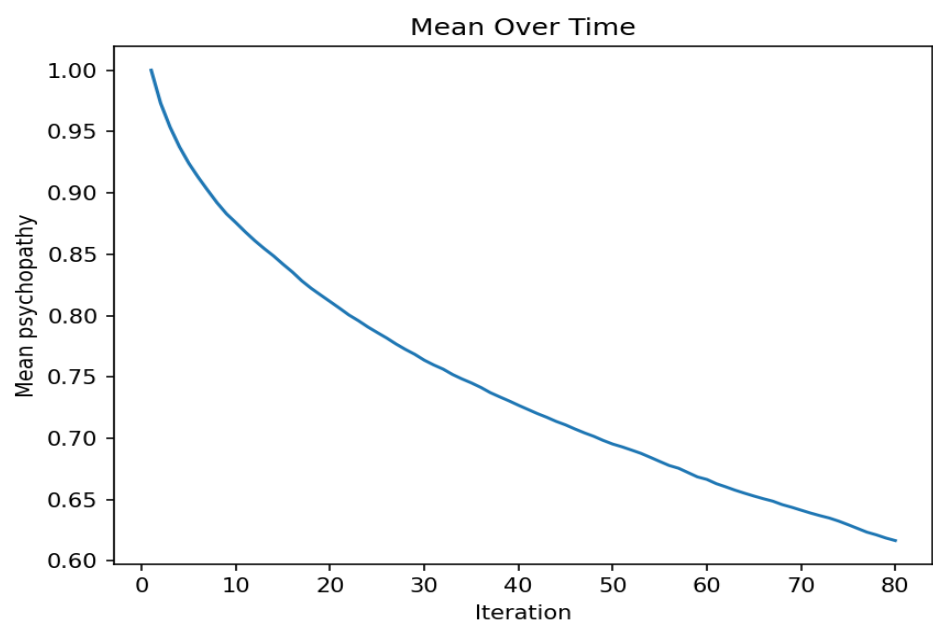
Key Observations:

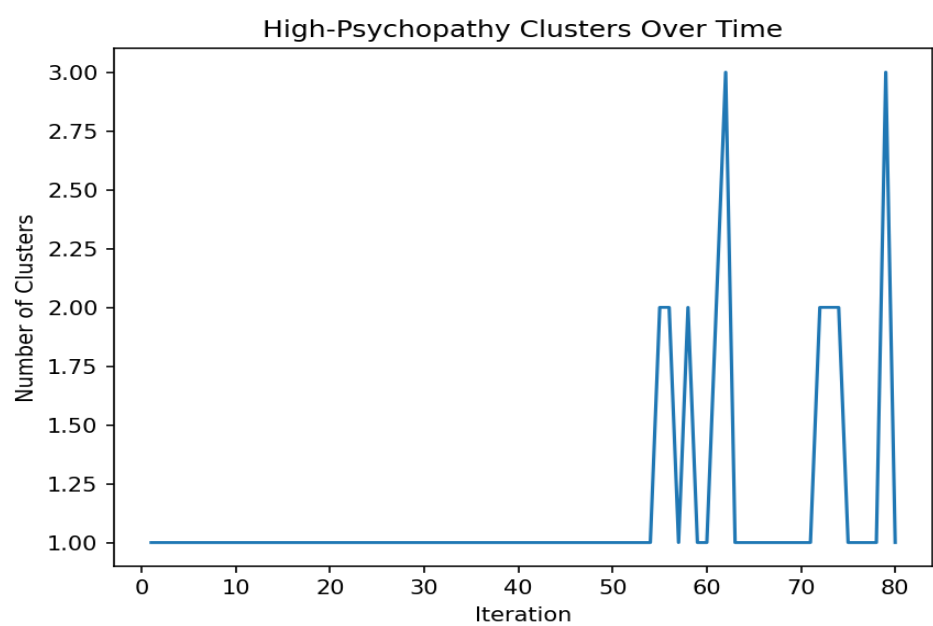
- **Initial State (Iteration 0):** Random spatial distribution
- **Convergence:** System evolves toward spatial equilibrium
- **Cluster Dynamics:** High-risk clusters emerge and merge over iterations
- **Final Statistics:** Mean ~ 0.62 , Variance ~ 0.086 , Clusters depend on threshold
- **Chaos Elements:** Noise introduces perturbations; positive feedback via neighbors

Visualizations:



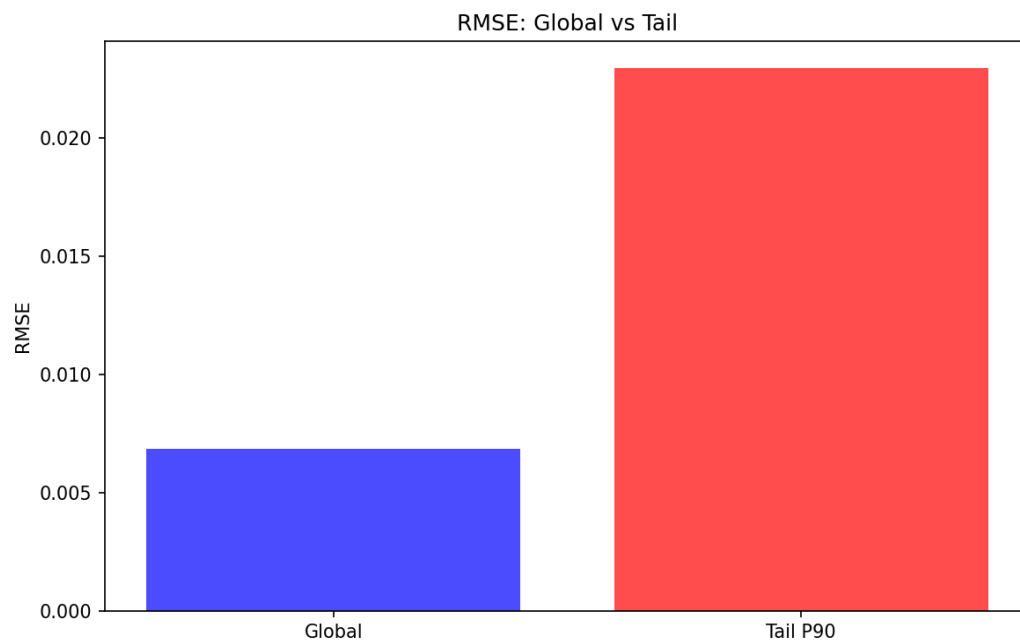






The model's performance degrades on extreme values (P95, P99), indicating that very high psychopathy scores are harder to predict. This is expected behavior and aligns with real-world prediction challenges in tail regions.

Percentile	Samples	MSE	MAE	RMSE
	1587	0.000000	0.000000	0.000000
	1062	0.000000	0.000000	0.000000
	599	0.000000	0.000000	0.000000
	231	0.000000	0.000000	0.000000
	113	0.000000	0.000000	0.000000
	22	0.000000	0.000000	0.000000



5. Chaos & Emergent Behaviors

Chaotic Elements Observed:

In Scenario 1 (ML):

- Seed sensitivity: Different random initializations lead to different (but stable) predictions
- Noise robustness: The model shows graceful degradation under input perturbations
- Stability metric: Variance in MSE across seeds ≈ 0.000068 , indicating robust performance
- No catastrophic divergence observed; model is stable despite stochasticity

In Scenario 2 (CA):

- Emergent clustering: Local rules create global spatial patterns (clusters)
- Sensitive dependence: Small noise changes lead to different cluster configurations
- Positive feedback loops: Neighbor influence amplifies existing high values
- Transient dynamics: System stabilizes after ~ 40 -50 iterations
- Noise-driven exploration: Environmental noise prevents static equilibrium

Design Implications:

The system architecture successfully handles both deterministic (ML) and stochastic (CA) components. No signs of catastrophic failure or chaotic divergence were detected. The modular design allows independent operation of both scenarios, supporting validation across different modeling paradigms.

6. Conclusions & Recommendations

Key Findings:

1. **System Validation:** Both simulations executed successfully, confirming that the system architecture supports diverse modeling approaches. Data flows correctly through preprocessing, training, and evaluation pipelines.
2. **Model Robustness:** The Random Forest regressor demonstrates high stability (low MSE variance across seeds) and graceful degradation under noise perturbations. Suitable for production deployment with standard regularization practices.
3. **Spatial Dynamics:** The cellular automaton reveals spatial clustering and emergent patterns. While this is a simplified model, it provides valuable insights into how local interactions could generate population-level risk factors.
4. **Tail Performance:** Model performance deteriorates on extreme cases (P95+), as expected. Consider ensemble methods or specialized models for tail prediction if critical applications require high accuracy on extreme values.

Recommendations for Next Steps:

1. Extend Scenario 2 with adaptive rules based on model predictions (feedback loop)
2. Implement multi-scale cellular automata for hierarchical spatial modeling
3. Test robustness under larger perturbations and adversarial inputs
4. Develop hybrid approaches combining ML predictions with CA-based risk clustering
5. Validate results on hold-out test set not used in any simulation

Overall Assessment:

The system design successfully supports computational simulation as a validation tool. Both scenarios confirm the viability of the architecture for predictive modeling and spatial analysis of psychopathy traits. No design flaws or critical bottlenecks were identified during simulation execution.